

Light Microscopy

Light microscopy operates by exposing a sample to light and positioning a series of lenses between the eye (or camera sensor) and that sample. These stacked sequences of lenses produce a magnification effect, depending on their spacing and curvature, due to the way that photons travel in curved glass. This same basic principle lies behind telescopes and camera lenses. In fact, your smartphone's camera, as small as it is, may have a dozen or more lenses carefully stacked behind the exterior lens you see!



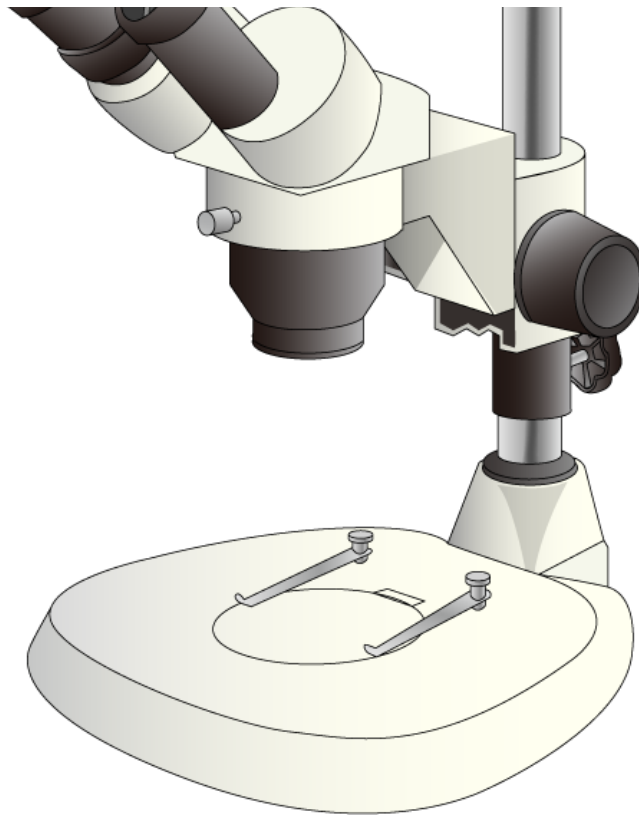
The stacked lenses of the iPhone 13 Pro Ultra-Wide Camera.

Source: [TechCrunch \(https://techcrunch.com/2021/09/14/apple-brings-macro-low-light-and-cinema-focused-updates-to-the-iphone-13-pro-camera/\)](https://techcrunch.com/2021/09/14/apple-brings-macro-low-light-and-cinema-focused-updates-to-the-iphone-13-pro-camera/)

Reflected Light Microscopy

Reflected light microscopes are often termed **stereomicroscopes**, due to their use of two light pathways, one to each eye, which permits depth perception. Such microscopes operate at low magnifications (generally less than 60x, but sometimes reaching 100x) but have increased depth of field (a greater depth is in focus at any one time) in comparison with transmitted light microscopes, which focus in a single plane at a time. Light is shone on the object of interest and reflects into the microscope's optics: hence, the name of the technique. Reflected light microscopy is typically used in archaeology to examine macroscopic botanical remains, such as seeds and wood charcoal.





An illustration of a stereomicroscope that uses reflected light to illuminate the sample. The sample would be placed on the stage where the clips are visible.

Source: [Wikimedia Commons](#)

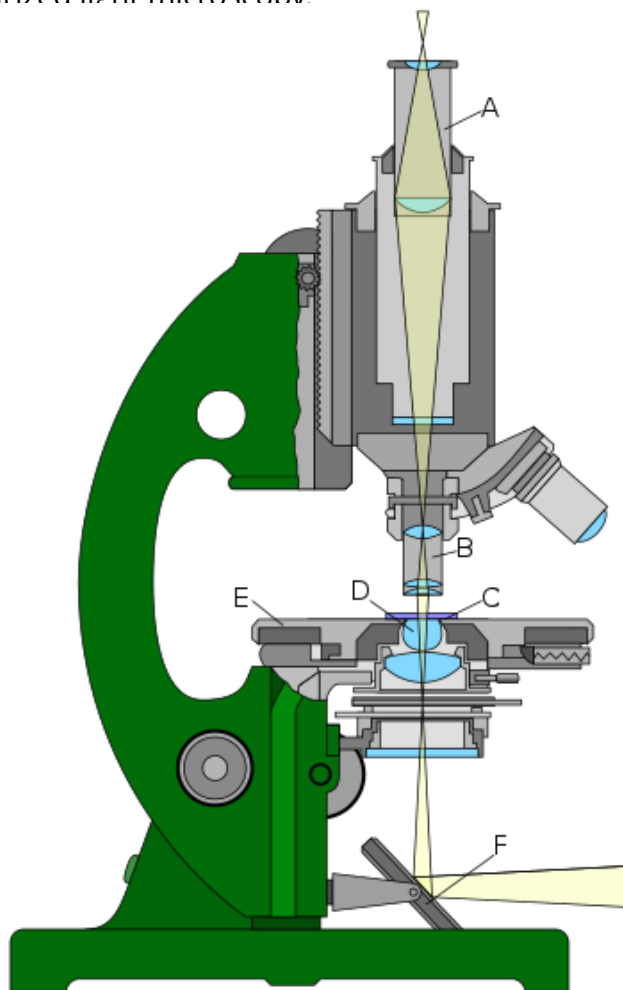
(https://commons.wikimedia.org/wiki/File:201103_stereomicroscope.png)

Stereomicroscopes are reflected light microscopes with two optical pathways, one of which reaches each eye. This allows greater depth perception of the sample than a single-eyepiece design.

If you are curious and want to learn more about methods of illumination in reflected light microscopy, check out Nikon's [Microscopy U](https://www.microscopyu.com/) (<https://www.microscopyu.com/>) website. It's a fantastic resource, and they have a great [article on illumination](https://www.microscopyu.com/techniques/stereomicroscopy/reflected-episcopic-light-illumination) (<https://www.microscopyu.com/techniques/stereomicroscopy/reflected-episcopic-light-illumination>) with excellent diagrams.

Transmitted Light Microscopy

Transmitted light microscopy relies on a simpler design with a single stack of lenses, as shown below. Light is shone from beneath the sample, passes through the sample (the light is transmitted through the sample: hence, the name), and continues through the lenses of the microscope to the eye. Transmitted light microscopy typically allows much higher levels of magnification (usually 20-500x) but a very limited depth of field, since only a single plane is in focus at a time. Samples must be able to be mounted on glass slides to be viewed; ideally, samples should also transmit light, at least partially. Archaeologists use transmitted light microscopy to view microscopic plant remains, especially phytoliths and pollen, and to take high-resolution images of wood, if that wood has been carefully thin-sectioned before being mounted on a glass slide for viewing. Starch granule identification requires polarized light microscopy.



Section illustration of a transmitted light microscope. Blue elements are lenses. C is the sample (on a glass slide); F is here a mirror but in other designs would be a light source in other designs.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Microscope-letters.svg)

(<https://commons.wikimedia.org/wiki/File:Microscope-letters.svg>)

Polarized Light Microscopy

Polarization is the phenomenon of restricting light to a single plane of movement. Cross-polarized light involves using two filters. The first ensures that only specific parallel wavelengths of light pass through the sample (initial polarization). That light is transmitted through the sample and passes through another polarizing filter oriented perpendicularly to the original filter. This blocks all light that has not been bent into another plane by passing through the sample; this is why the background color changes from light to dark. Only the samples with refringent properties that have changed the plane of transmitted light appear in the image.

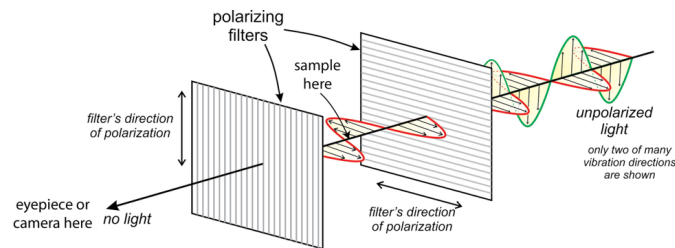


Diagram of cross-polarized light. The light source (right) emits light that moves at multiple angles; the first (polarizing) filter reduces this light to that moving in a single plane. That light passes through a sample and to a second, perpendicular polarizing filter. Only light that has interacted with the sample and changed planes will be transmitted through the second filter into the eyepiece or camera.

Source: Modified from (<https://opengeology.org/Mineralogy/5-optical-mineralogy/>)
OpenGeology's Introduction to Mineralogy, Ch. 5
(<https://opengeology.org/Mineralogy/5-optical-mineralogy/>)

Polarization describes the phenomenon in which waves of light can be restricted to a single plane of movement or oscillation. Since light can be described as a wave, polarized light is light in which the waves are limited to one plane.

A polarized light microscope is a transmitted light microscope with the addition of two polarizing filters: one between the light source and the sample (between F and C, in the transmitted light microscope diagram above), and the second, perpendicular polarizing filter between the sample and the eyepiece (between C and A, in the transmitted light microscope diagram above). Polarized light is particularly valuable in identifying starch granules, which are birefringent and illuminate in specific ways during examination with cross-polarized filters.